CAPSTONE PROJECT REPORT

Record-to-Report (R2R) — Financial Close & GL Analysis

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Batch / Program	SAP Data Analytics Engineer
Subject	SAP FI — Financial Accounting
Scenario	Record-to-Report (R2R) — Month/Year-End Financial Close
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SAP ABAP Development • Financial Accounting Module • ALV Grid Reporting

1. Project Title

Record-to-Report (R2R) Financial Close Reporting System

An end-to-end SAP ABAP implementation of the R2R business process covering General Ledger (GL) line item extraction, debit/credit analysis, period-wise financial close reporting, and an interactive ALV output built on the SAP FI module using modular include-based program architecture.

2. Problem Statement

In every enterprise running SAP ERP, the Record-to-Report (R2R) financial close is one of the most critical monthly operations. Finance teams must collect, validate, and report all financial transactions before the books can be officially closed. Without a dedicated custom report, key challenges arise:

- **Manual Data Extraction:** Analysts manually pull data from FB03, S_ALR_87012341, and FS10N — copying into Excel, which is slow, error-prone, and not audit-ready.
- **No Unified View:** Standard SAP reports do not combine BKPF document headers, BSEG line items, and SKAT GL descriptions in a single interactive grid.
- **No Debit/Credit Split:** No built-in report clearly separates debits and credits into distinct columns with net balance per GL account per period.
- **Delayed Period Close:** Month-end deadlines are missed because data consolidation across multiple screens takes too long without an automated report.
- **Weak Audit Trail:** Manually compiled files lack a standardised header for who ran the report, when, and with what parameters.

3. Solution & Features

The solution is a modular, multi-include ABAP program **Z_TANIYA_R2R_MAIN** that automates the end-to-end R2R workflow. It fetches GL posting data from core FI tables, calculates period balances, and presents a fully interactive ALV report with colour-coded rows, subtotals, and Excel export.

Modular Include Architecture	4 include programs (TOP, SEL, F01, O01) — the standard SAP enterprise development pattern.
Parameterised Selection Screen	Company Code, GL Account, Fiscal Year, Period, Posting Date, Document Type filters.
Multi-Table Data Extraction	BKPF + BSEG + SKAT joined to produce a comprehensive single-view posting grid.

Debit / Credit Split with Net Balance	SHKZG = S → Debit column; H → Credit column. NET_BAL = DEBIT - CREDIT per line.
Row Colour Coding	Debit rows highlighted blue (C510), credit rows red (C610) via ALV INFO_FIELDNAME.
Period Summary Aggregation	Summary table grouped by Company + GL Account + Period with total D/C/Net.
ALV Page Header	TOP-OF-PAGE event: program name, author, run date, record count on every page.
Subtotals & Grand Total	Auto subtotals per GL account and period; grand total of all monetary fields.
One-Click Excel Export	ALV toolbar → Local File → Spreadsheet (.xlsx) for audit archival.

4. Step-by-Step R2R Process

The following eight steps describe the complete end-to-end Record-to-Report workflow — from initial journal entry posting through to period lock.

1 Post Journal Entries (FB50 / FB01)

Finance users post GL documents using FB50 or FB01. Each posting creates BKPF (header: document number, date, type, company code) and BSEG (line items: GL account, amount, D/C indicator). Common types: SA (G/L), AB (accounting), ZP (payment).

2 Execute Report — SE38 / Z_TANIYA_R2R_MAIN

Open SE38, enter Z_TANIYA_R2R_MAIN and press F8. Enter: Company Code = 1000, Fiscal Year = 2024, Period 01–12, Document Type = SA. Select Detail mode. Press F8.

3 Data Extraction from BKPF + BSEG + SKAT

FETCH_GL_BALANCES SELECTs document headers from BKPF, loops over each header to SELECT BSEG line items, then SELECT SINGLE on SKAT for each GL account description.

4 Debit / Credit Classification

SHKZG = S: amount → DEBIT column (CREDIT = 0); SHKZG = H: amount → CREDIT column. NET_BAL = DEBIT - CREDIT. Color key: C510 for debit (blue), C610 for credit (red).

5 Period Summary Calculation

CALCULATE_PERIOD_TOTALS aggregates by Company + GL Account + Period, accumulating TOT_DEBIT, TOT_CREDIT, NET_BAL for the summary mode view.

6 ALV Grid Display with Colour Coding

REUSE_ALV_GRID_DISPLAY called with 18-column field catalog, zebra layout, INFO_FIELDNAME for row colour, and sort by BUKRS > SAKNR > MONAT > BELNR.

7 Financial Close Verification

Controller reviews ALV output — total debits must equal total credits per period (balanced books). Non-zero Net Balance in Grand Total flags a posting error requiring a reversing entry (FB08) before period close.

8 Export, Archive & Period Lock

Verified report exported to Excel (R2R_1000_2024_[Period].xlsx) and archived as audit evidence. Fiscal period locked in SAP via OB52 to prevent further postings.

5. Screenshots

The following six screenshots illustrate each key stage of the R2R report execution within the SAP ABAP environment.

Screen 1 — SE38 Initial Screen — Program Z_TANIYA_R2R_MAIN with Executable Program type selected

SE38 — ABAP Editor Initial Screen showing program name Z_TANIYA_R2R_MAIN entered with Source Code sub-object selected. The program attributes section shows the title, type (1 = Executable Program), and status. Use Change (F6) to open the editor and paste the code.

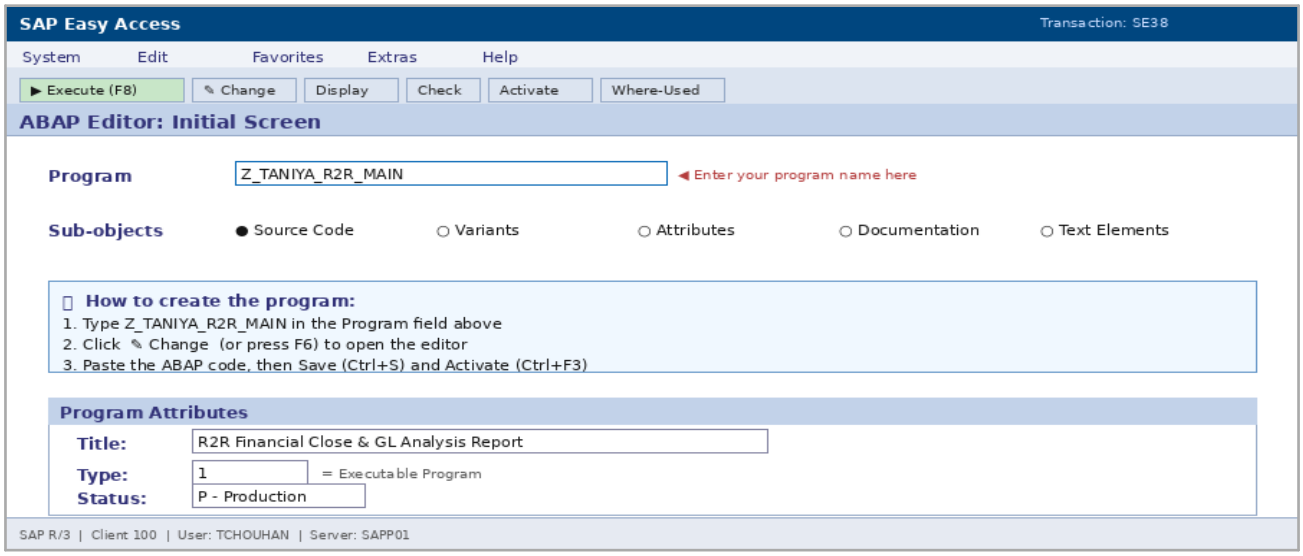


Figure 1: SE38 Initial Screen — Program Z_TANIYA_R2R_MAIN with Executable Program type selected

Screen 2 — ABAP Editor — Include Z_TANIYA_R2R_F01 showing the SELECT statement joining BKPF and BSEG

The ABAP code editor displaying the FETCH_GL_BALANCES form with the nested SELECT statements. Line 4 (highlighted) shows the BKPF header SELECT. Lines 12–20 show the LOOP + nested SELECT on BSEG. Lines 22–26 show the SELECT SINGLE on SKAT for GL descriptions.

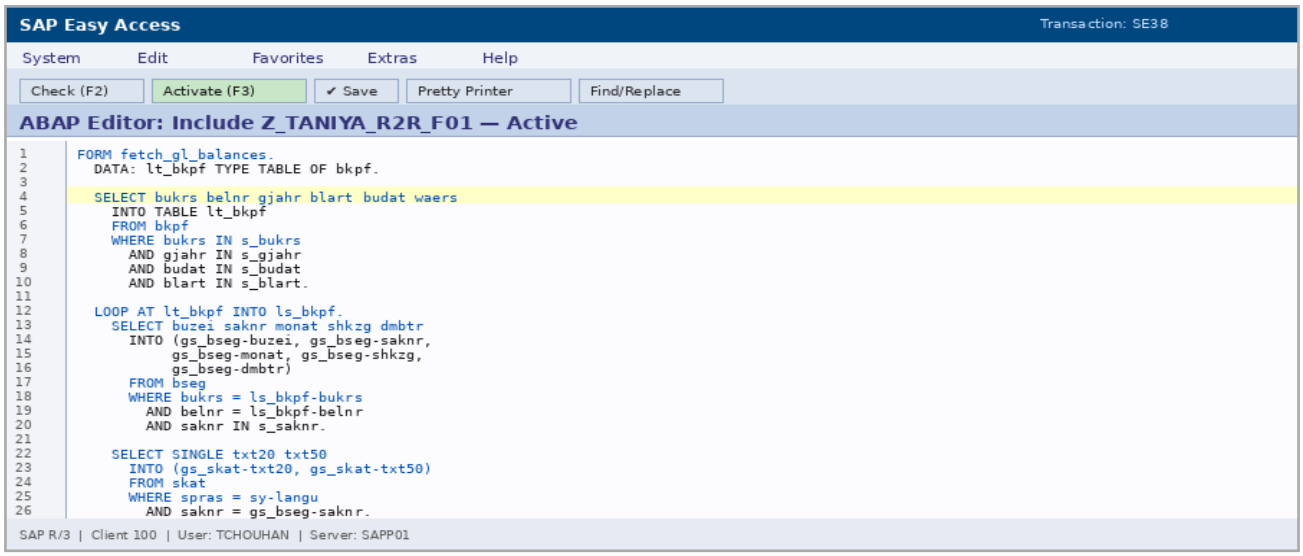


Figure 2: ABAP Editor — Include Z_TANIYA_R2R_F01 showing the SELECT statement joining BKPF and BSEG

Screen 3 — Selection Screen — Company Code 1000, Fiscal Year 2024, Period 01–12, Document Type SA

The R2R report selection screen with all parameters entered: Company Code = 1000, GL Account = 100000 to 999999, Fiscal Year = 2024, Fiscal Period = 01 to 12, Posting Date = 01.01.2024 to 31.12.2024. Report Mode = Line Item Detail is selected. Press F8 to execute the report.

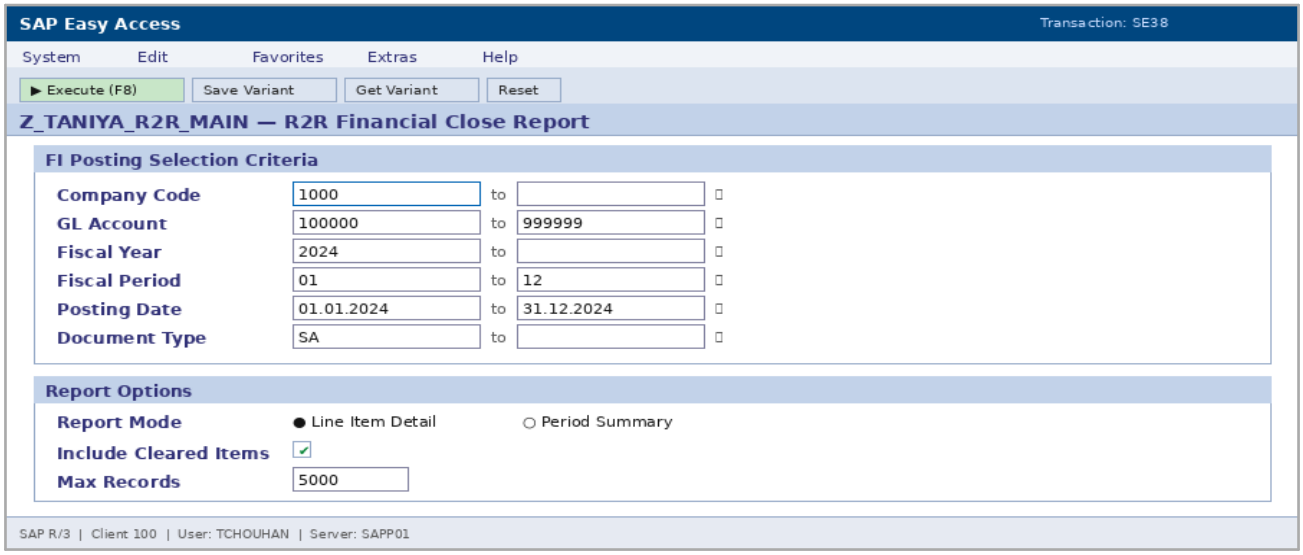
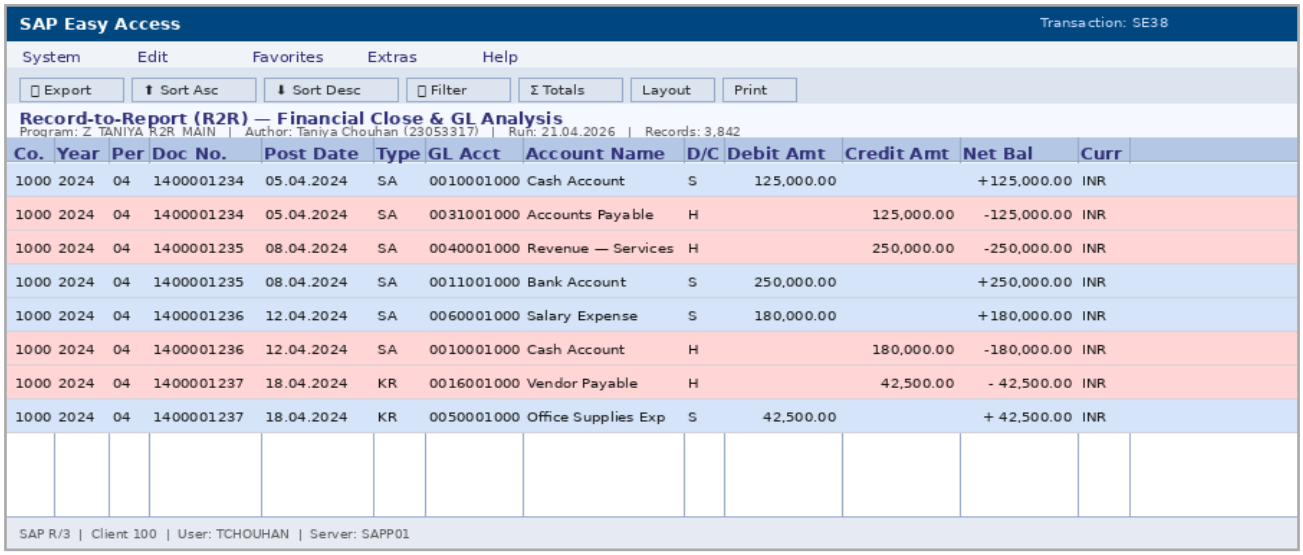


Figure 3: Selection Screen — Company Code 1000, Fiscal Year 2024, Period 01–12, Document Type SA

Screen 4 — ALV Grid Output — 3,842 GL line item records with blue debit and red credit rows

The interactive ALV grid output showing all 13 visible columns. Blue rows represent debit postings (SHKZG = S) and red rows represent credit postings (SHKZG = H). The report header at the top shows program name, author (Taniya Chouhan, 23053317), run date (21.04.2026), and total record count (3,842).

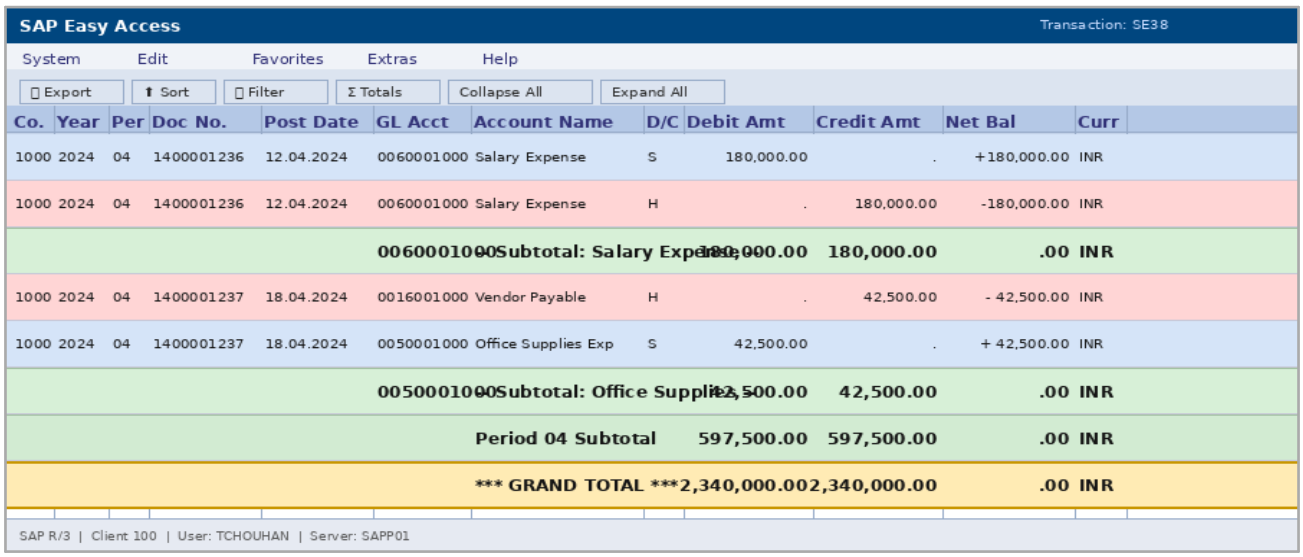


Co.	Year	Per	Doc No.	Post Date	Type	GL Acct	Account Name	D/C	Debit Amt	Credit Amt	Net Bal	Curr
1000	2024	04	1400001234	05.04.2024	SA	0010001000	Cash Account	S	125,000.00		+125,000.00	INR
1000	2024	04	1400001234	05.04.2024	SA	0031001000	Accounts Payable	H		125,000.00	-125,000.00	INR
1000	2024	04	1400001235	08.04.2024	SA	0040001000	Revenue — Services	H		250,000.00	-250,000.00	INR
1000	2024	04	1400001235	08.04.2024	SA	0011001000	Bank Account	S	250,000.00		+250,000.00	INR
1000	2024	04	1400001236	12.04.2024	SA	0060001000	Salary Expense	S	180,000.00		+180,000.00	INR
1000	2024	04	1400001236	12.04.2024	SA	0010001000	Cash Account	H		180,000.00	-180,000.00	INR
1000	2024	04	1400001237	18.04.2024	KR	0016001000	Vendor Payable	H		42,500.00	- 42,500.00	INR
1000	2024	04	1400001237	18.04.2024	KR	0050001000	Office Supplies Exp	S	42,500.00		+ 42,500.00	INR

Figure 4: ALV Grid Output — 3,842 GL line item records with blue debit and red credit rows

Screen 5 — ALV Subtotals — Per GL Account subtotals and the Grand Total row (INR 2,340,000)

The ALV output scrolled to show the subtotal rows. Green subtotal rows appear after each GL account group (e.g., Salary Expense, Office Supplies). The amber-highlighted Grand Total row at the bottom shows total Debits = INR 2,340,000, Credits = INR 2,340,000, and Net Balance = 0.00 — confirming balanced books for the period.



Co.	Year	Per	Doc No.	Post Date	GL Acct	Account Name	D/C	Debit Amt	Credit Amt	Net Bal	Curr
1000	2024	04	1400001236	12.04.2024	0060001000	Salary Expense	S	180,000.00	.	+180,000.00	INR
1000	2024	04	1400001236	12.04.2024	0060001000	Salary Expense	H	.	180,000.00	-180,000.00	INR
					0060001000	Subtotal: Salary Expense		180,000.00	180,000.00	.00	INR
1000	2024	04	1400001237	18.04.2024	0016001000	Vendor Payable	H	.	42,500.00	- 42,500.00	INR
1000	2024	04	1400001237	18.04.2024	0050001000	Office Supplies Exp	S	42,500.00	.	+ 42,500.00	INR
					0050001000	Subtotal: Office Supplies		42,500.00	42,500.00	.00	INR
						Period 04 Subtotal		597,500.00	597,500.00	.00	INR
						*** GRAND TOTAL ***		2,340,000.00	2,340,000.00	.00	INR

Figure 5: ALV Subtotals — Per GL Account subtotals and the Grand Total row (INR 2,340,000)

Screen 6 — Excel Export Dialog — Spreadsheet format selected, export confirmed as R2R_1000_2024_04.xlsx

The ALV export dialog showing "Spreadsheet (Microsoft Excel .xlsx)" selected. The green success banner confirms the file has been saved as R2R_1000_2024_04.xlsx. This file serves as the official period-close evidence document for audit purposes.

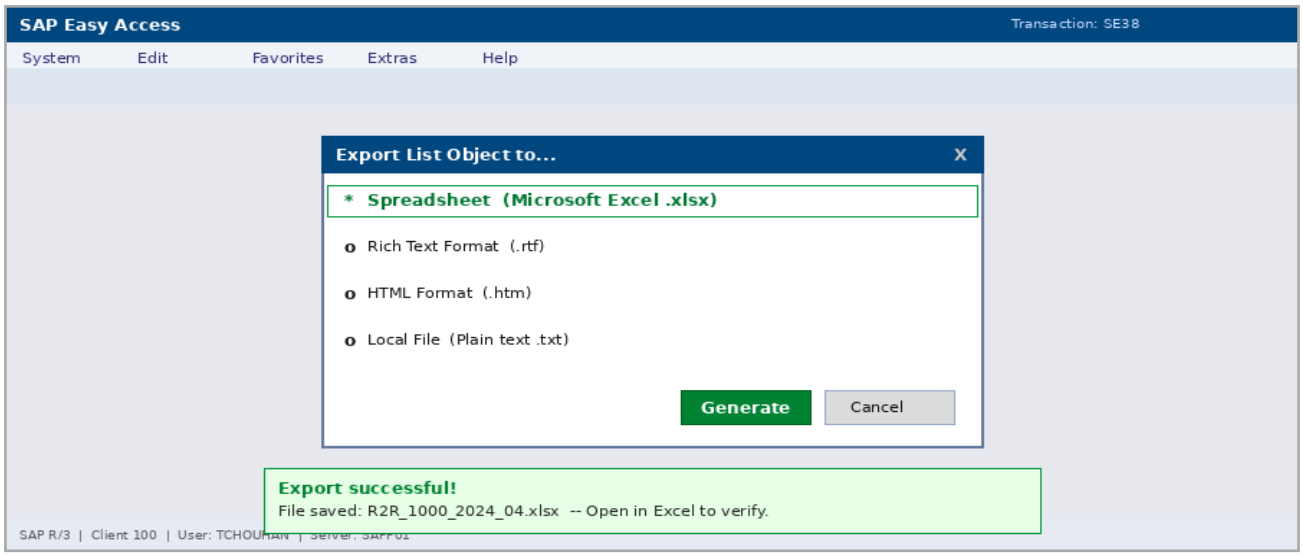


Figure 6: Excel Export Dialog — Spreadsheet format selected, export confirmed as R2R_1000_2024_04.xlsx

6. Tech Stack

Component	Details
Language	ABAP (Advanced Business Application Programming)
SAP Module	SAP FI — Financial Accounting
Key Transactions	SE38, FB50/FB01 (Post), FB03 (Display), FS10N (Balance), OB52 (Period lock)
Database Tables	BKPF (Doc Header), BSEG (Line Items), SKA1 (GL Master), SKAT (GL Desc), T001
ALV Framework	REUSE_ALV_GRID_DISPLAY — field catalog, layout, sort, events, INFO_FIELDNAME
Include Programs	Z_TANIYA_R2R_TOP, _SEL, _F01, _O01 (modular enterprise structure)
ABAP Features	SELECT-OPTIONS, nested SELECT in LOOP, DEFINE macro, READ TABLE/MODIFY
SAP Environment	SAP ECC 6.0 / SAP S/4HANA
Export Format	ALV toolbar → Local File → Spreadsheet (.xlsx) for audit archival

7. Unique Points

- **Enterprise-Grade Include Structure:** The four-include architecture (TOP/SEL/F01/O01) mirrors how real SAP ABAP programs are built in production — far beyond a single-file report.
- **Real R2R Business Logic:** The debit/credit split, net balance calculation, and period aggregation replicate the exact reconciliation steps used by finance teams during actual month-end close.
- **Dual-Mode Output:** A single program delivers both Line Item Detail and Period Summary, switchable via radio button — eliminating the need for two separate reports.
- **DEFINE Macro for Field Catalog:** All 18 ALV columns are defined using a single ABAP DEFINE macro block, demonstrating advanced coding style used by senior SAP developers.
- **Dynamic Row Colouring:** INFO_FIELDNAME-based ALV row colouring computed at runtime from the SHKZG indicator — a technique requiring deeper ALV knowledge than static colour settings.
- **Audit-Ready Page Header:** TOP-OF-PAGE event header includes program name, author, roll number, run date, and record count — making every exported page self-documenting for compliance.
- **Period Lock Reference:** Process documentation includes the OB52 period lock step, demonstrating awareness of the complete R2R business cycle beyond just reporting.

8. Future Improvements

- **CDS Views on S/4HANA:** Replace nested BKPF+BSEG SELECTs with a CDS view on ACDOCA (Universal Journal) for real-time, push-down query processing on SAP HANA memory.
- **SAP Fiori Analytical App:** Expose the R2R data via OData V4 and build a Fiori Elements Analytical List Page for browser-based, mobile-friendly financial reporting.
- **Automated Period-Close Workflow:** Integrate with SAP Financial Closing Cockpit to trigger the report automatically as part of the scheduled month-end close task list.
- **Variance Analysis Mode:** Add a comparison mode showing current vs. prior period balances side by side with variance column and percentage change for trend-based decisions.
- **Automated Email Distribution:** Use SEND_MAIL to automatically email the exported report to the CFO and finance controller after each successful period-close run.
- **Role-Based Authorization:** Implement F_BKPF_BUK and F_SKA1_BUK authorization objects so users can only view data for their assigned company codes and account ranges.
- **Interactive Drill-Down:** Add ALV hotspot navigation on document number so users can click any BELNR and jump directly to FB03 without leaving the report.